

Star Evolution Notes

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1 Observational background and basic assumptions

1.1 Introduction

A star to be a star must satisfy two conditions:

- It is bound by self-gravity (hence spherically symmetric)
- It radiates energy from an internal source

Hence the end of a star happens when one of these conditions fails: the star either isn't bound by its own gravity, and thus breaks up and scatters material, or it doesn't produce energy internally, meaning its nuclear fuel exhausts. Most stars end their lives in a combination of these two.

Primary characteristic that can be measured is the *apparent brightness* I_{obs} - is the flux of stars radiation (power over area), note that it's apparent since it depends on the distance to the observer. The power (energy per time) which radiated by a star is called *luminosity* L , so

$$I_{\text{obs}} = \frac{L}{4\pi d^2}$$

where d is the distance to the observer. The effective temperature of a star T_{eff} is thus defined as the temperature of a blackbody. This is an approximation for the star's outermost layer - *photosphere* (bulk of the emitted radiation comes from here).

$$\sigma T_{\text{eff}}^4 = \frac{L}{4\pi R^2}$$

where σ is the Stefan-Boltzman constant. According to Wein's law the maximum radiation wavelength λ_{max} is given by

$$\lambda_{\text{max}} T = \text{const}$$

Based on evidence there are some axioms that can be added to the general definition of a star

- **Isolation** A single star may be considered isolated in empty space. Hence, initial conditions will exclusively determine its evolution.
- **Uniform initial composition.** There is little difference in the initial composition of stars, of which the abundance of heavy elements within a star have the biggest influence, which is negligible compared to hydrogen and helium. The fate of the star is then solely dependent upon its initial mass M .
- **Spherical Symmetry** In vast majority of the cases the rotation of a star or its magnetic field are not strong enough to lose spherical symmetry. Therefore, the physical properties of a star change with just the radius r .

$$dm = \rho 4\pi r^2 dr \quad \Rightarrow \quad m(r) = \int_0^r 4\pi r^2 \rho(r) dr$$

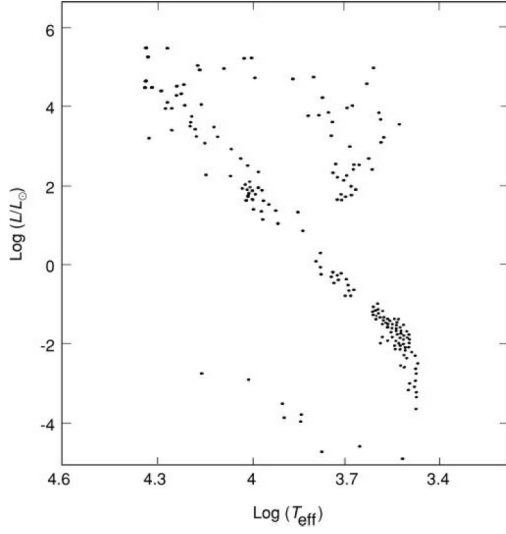
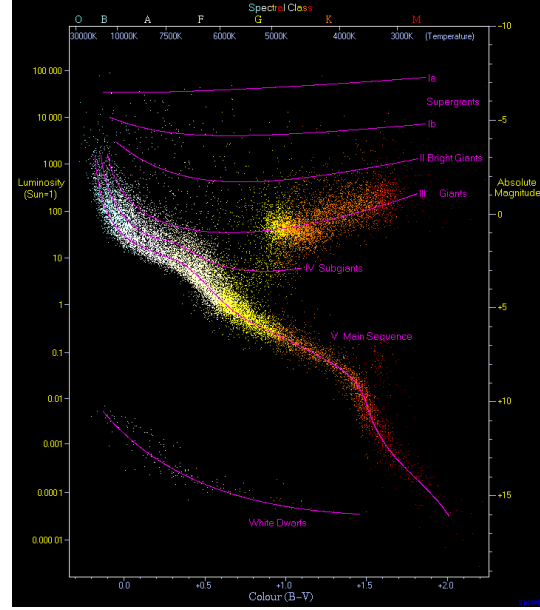


Figure 1.3 The H-R diagram of stars in the neighbourhood of the Sun.



The thin strip in the middle is called **the main sequence** which corresponds stars called **main sequence stars**. The right blob above the main sequence (has the same or lower T_{eff} at the same L) have their spectrum shifter toward the longer wavelengths with a reddish colour. According to equations higher L and lower T_{eff} implies a large radius. Such stars are called **red giants**. Their radii may attain several hundred solar radii.

Conversely, on the lower left corner are stars with low luminosities and high effective temperatures. Thus they have small radius and a bluish-white colour. They are accordingly named **white dwarfs**. Their radii are about the size of Earth's but they have the mass of the sun.

From the axioms we know that initial conditions (mass and age) are the main differentiators. Hence there are two ways to interpret HR diagram: either the stars in the diagram represent stars at different ages, thus allowing us to draw a line and seeing how the star evolves, or it's properties (Luminosity and Surface temperature) are highly dependent on mass. In general, we can conclude that being outside the main sequence is determined by age, while the location *in main sequence* depends on mass. Note that the hotter the star in the main sequence the more massive it is. In addition, since high/low mass stars burn fuel way faster/slower, their lifetimes in main sequence are shorter/longer. Hence the lower part of the main sequence (low temperature and mass) is basically constantly populated. In fact, there is a power-law dependence of main-sequence stars

$$L \propto M^\nu \quad \frac{L}{L_\odot} = \left(\frac{M}{M_\odot} \right)^\nu$$

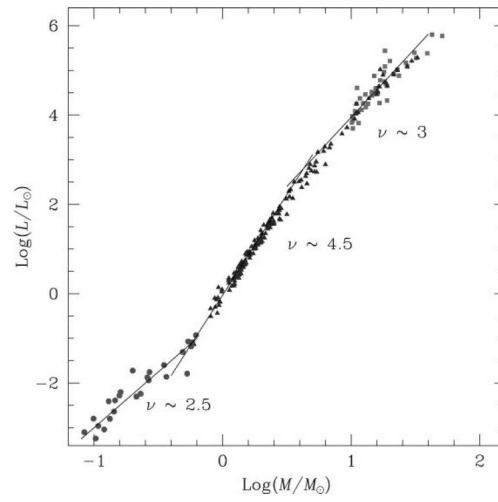


Figure 1.6 The mass-luminosity relation for main-sequence stars. Data from O. Yu. Malkov

2 The equations of stellar evolution

Start of with the mass element

$$dm = \rho dV = \rho 4\pi r^2 dr$$

From the first law of thermodynamics (assuming the conservation of mass with $dm \equiv \text{const}$)

$$\delta(udm) = dm\delta u = \delta Q + \delta W$$

where u is the internal energy per unit mass. We can calculate the work δW

$$\delta W = -P\delta V = -P\delta\left(\frac{dV}{dm}dm\right) = -P\delta\left(\frac{1}{\rho}\right)dm$$

Next, note that heat comes from two main sources: release of nuclear energy and the balance of the heat fluxes, going in and out. The rate of nuclear energy per unit mass is q and The heat flowing though the surface is $F(m)$ (units of power), and clearly $F(M) = L$. Thus for heat

$$\delta Q = qdm\delta t + F(m)\delta t - F(m+dm)\delta t$$

Using $F(m+dm) = F(m) + \frac{\partial F}{\partial m}dm$, leaves us with

$$\delta Q = \left(q - \frac{\partial F}{\partial m}\right)dm\delta t$$

Inserting everything back into first law we obtain

$$dm\delta u + P\delta\left(\frac{1}{\rho}\right)dm = \left(q - \frac{\partial F}{\partial m}\right)dm\delta t$$

It seems that dm cancels and taking the limit $\delta t \rightarrow 0$ we get

$$\dot{u} + P\left(\frac{\dot{1}}{\rho}\right) = q - \frac{\partial F}{\partial m}$$

where the dot is a **partial** derivative wrt time. In thermal equilibrium there is no change in these quantities wrt time, hence the time derivatives vanish

$$q = \frac{dF}{dm} \Rightarrow \int_0^M qdm = \int_0^M dF = L$$

This integration gives the total power supplied by nuclear processes, denoted by L_{nuc} as **nuclear luminosity**. Thermal equilibrium implies that the energy created is radiated away at the same rate (No energy build up).

Next we can derive the equations of motion. Start with some mass

$$\Delta m = \rho dr dS$$

Any piece of mass feels two opposing forces: gravitational force and force from pressure. Due to spherical symmetry we can only consider the radial part of the acceleration

$$\ddot{r}\Delta m = -\frac{Gm\Delta m}{r^2} + P(r)dS - P(r+dr)dS$$

Again, $P(r + dr) = P(r) + \frac{\partial P}{\partial r} dr$, so

$$\ddot{r} \Delta m = -\frac{Gm \Delta m}{r^2} - \frac{\partial P}{\partial r} \frac{\Delta m}{\rho} \Rightarrow \ddot{r} = \frac{Gm}{r^2} - \frac{1}{\rho} \frac{\partial P}{\partial r}$$

Then using $dr = dm/(4\pi r^2 \rho)$

$$\ddot{r} = -\frac{Gm}{r^2} - 4\pi r^2 \frac{\partial P}{\partial m}$$

When acceleration is negligible we get **hydrostatic equilibrium**

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2} \quad \text{or} \quad \frac{dP}{dm} = -\frac{Gm}{4\pi r^4}$$

We can estimate the pressure by integrating from the center to the outside

$$P(M) - P(0) = -\int_0^M \frac{Gm dm}{4\pi r^4} \Rightarrow P_c = \int_0^M \frac{Gm dm}{4\pi r^4} > \int_0^M \frac{Gm dm}{4\pi R^4}$$

2.1 Virial Theorem

Multiplying the hydrostatic equilibrium equation by volume V

$$\int_0^{P(R)} V dP = -\frac{1}{3} \int_0^M \frac{Gm dm}{r}$$

But the RHS is just the gravitational potential energy Ω and we can integrate LHS by parts where boundary terms vanish since $V(0) = 0$ and $P(R) = 0$. Yielding

$$-3 \int_0^{V(R)} P dV = \Omega \Leftrightarrow -3 \int_0^M \frac{P}{\rho} dm = \Omega$$

We can use this result, for ideal gas.

$$u = \frac{3}{2} \frac{kT}{m_g} = \frac{3}{2} \frac{P}{\rho}$$

Then combining these equations and noting that $U = \int u dm$ we get

$$U = -\frac{1}{2} \Omega$$

We can use these formulas to derive average temperature. For $\Omega = -\alpha \frac{GM^2}{R}$ and $U = \frac{3}{2} \frac{k}{m_g} \bar{T} M$ we obtain

$$\bar{T} = \frac{\alpha}{3} \frac{m_g G}{k} \frac{M}{R}$$

2.2 Total Energy of the star

- Parsec (pc) - distance at which 1AU subtends an angle of 1 arcsecond
- Parallax - the apparent shift of a nearby star against distant background stars due to Earth's orbital motion. Used to measure stellar distances
- Proper motion - the angular motion of a star across the sky (per year), excluding parallax
- Luminosity - the total energy output of an object per unit time
- Brightness (Flux) - How bright an object appears from Earth

$$F = \frac{L}{4\pi d^2}$$

- Magnitude - a logarithmic measure of brightness:
 - Apparent magnitude (m) - how bright the object looks from Earth

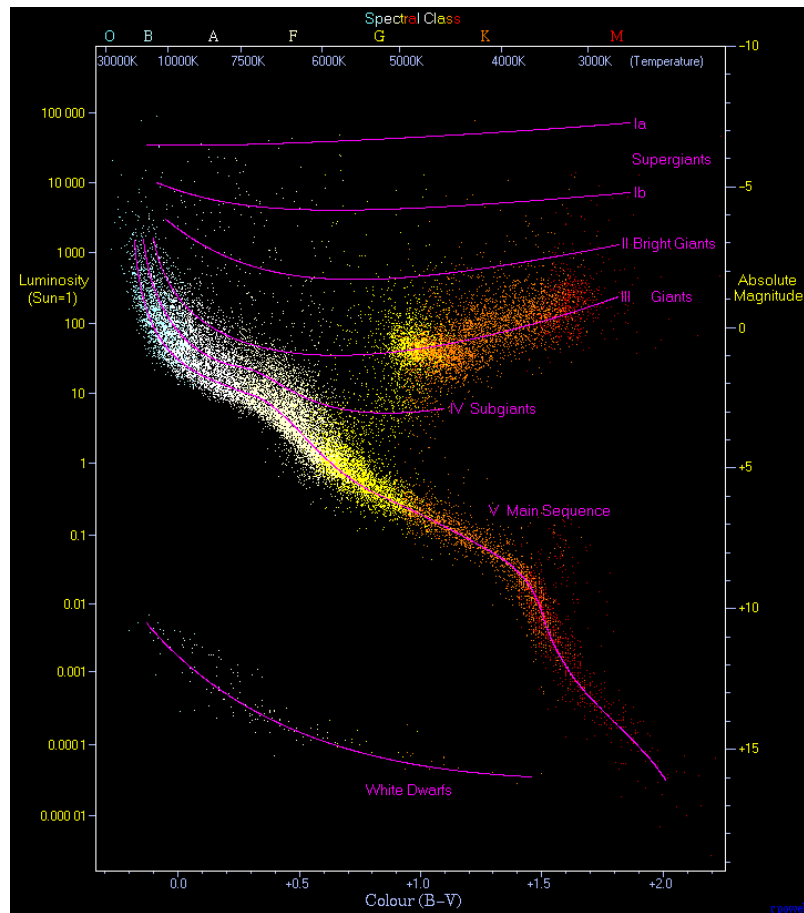
$$m_2 - m_1 = -2.512 \log \left(\frac{F_2}{F_1} \right)$$

- absolute magnitude (M) - brightness at a standard distance of 10 pc
- Effective temperature (T_{eff}) Defined via $F = \sigma T_{\text{eff}}^4$. Temperature that a blackbody would emit the same total radiation as the star. It determines stellar color and spectral type.
- Spectral Class - Classification of stars by their spectra and temperature:

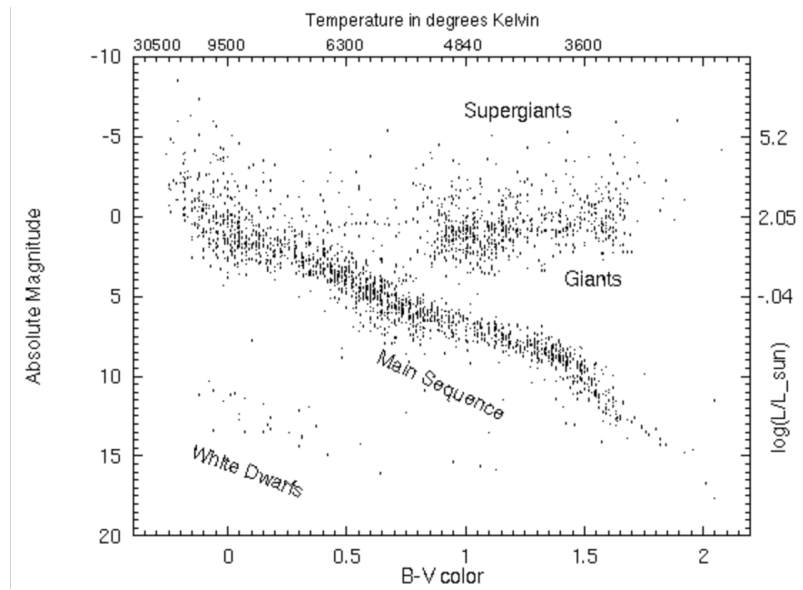
O-B-A-F-G-K-M

from hottest to coolest. Reflects surface temperature and ionization states of elements. mnemonic "Oh be a fine girl, kiss me"

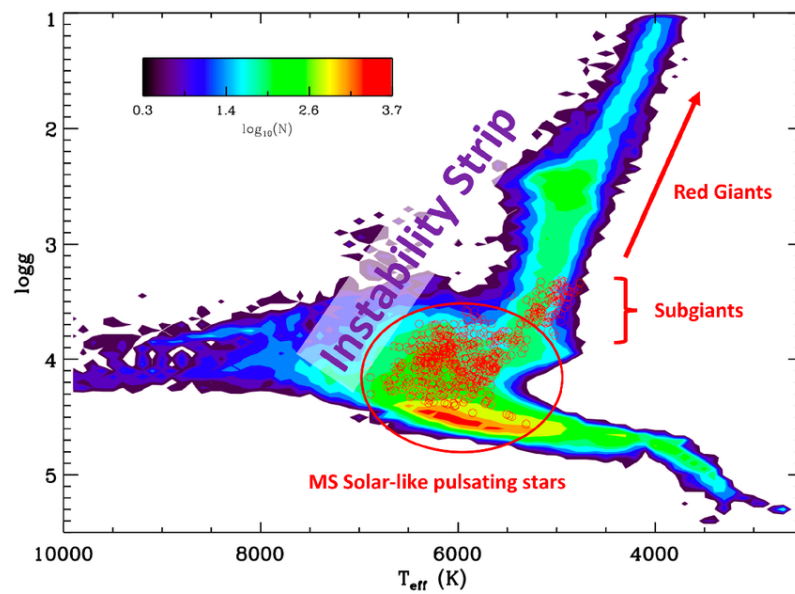
- Hertzsprung-Russell diagram (HR-diagram) - a plot of luminosity vs temperature (of spectral class)



- Colour-Magnitude diagram (CM-diagram) - observational version of HRD (magnitude vs color index). Used heavily for star-clusters, since distance is common



- Kiel Diagram - a theoretical evolution diagram (effective temperature vs surface gravity ($\log g$))



- Kippenhahn diagram - a time-interior diagram of a star: shows convective vs radiative zones, nuclear burning regions, evolution of stellar structure over time

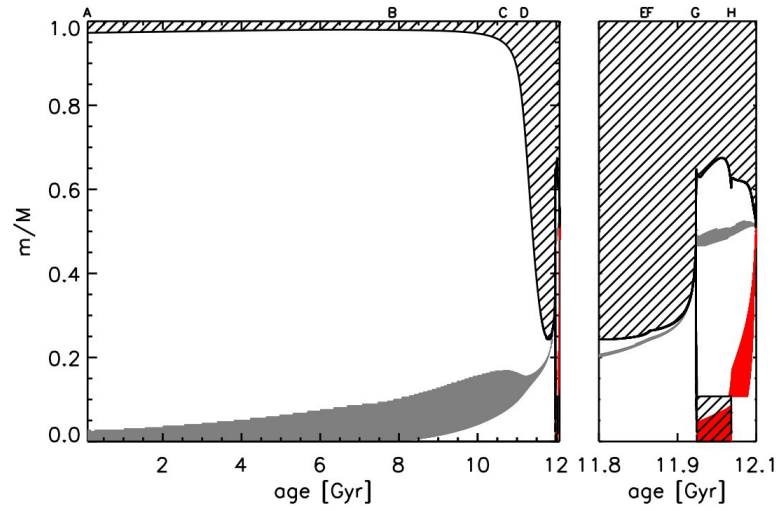


Fig. 4 Kippenhahn diagram of a $1 M_{\odot}$ model shown in Fig. 3. Regions where convection takes place are hatched. Regions where nuclear burning produces more than $10 \text{ erg g}^{-1} \text{ s}^{-1}$ are shown in grey for hydrogen burning and in red in case of helium burning. The right panel shows a zoom of the late stages of evolution. The letters at the top indicate different stages of evolution corresponding to the phases indicated in Fig. 3.

- T_c - ρ_c diagram - a plot of central temperature T_c vs central density ρ_c . Tracks internal evolution of a star, especially during late stages.

